

Implementasi Random Forest yang Telah Dilatih Pada Perangkat ESP32

Menggunakan File Header (.h)

```
#pragma once
```

```
#include <stdint.h>
```

```
namespace Eloquent::ML::Port {  
    class RandomForest {  
    public:  
        /*  
        ** Predict class for feature's vector  
        */  
        static int predict(const float *x) {  
            uint8_t votes[2] = {0};  
            // tree #1  
            if (x[0] <= 681.8073425292969) {  
                if (x[1] <= 34.89121437072754) {  
                    votes[1] += 1;  
                } else {  
                    if (x[2] <= 44.45821762084961) {  
                        votes[1] += 1;  
                    } else {  
                        votes[1] += 1;  
                    }  
                }  
            }  
            } else {  
                votes[0] += 1;  
            }  
            // tree #2  
            if (x[2] <= 38.432024002075195) {
```

```

if (x[2] <= 38.02525329589844) {
    votes[1] += 1;
} else {
    if (x[2] <= 38.22780418395996) {
        votes[0] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[2] <= 80.87892150878906) {
        if (x[1] <= 38.37708854675293) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 660.0310974121094) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}
// tree #3
if (x[1] <= 29.751760482788086) {
    if (x[1] <= 29.3792667388916) {
        if (x[1] <= 29.097238540649414) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

    }
} else {
    if (x[0] <= 664.5515441894531) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[2] <= 38.467233657836914) {
        votes[0] += 1;
    } else {
        if (x[2] <= 43.26704978942871) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}
}
// tree #4
if (x[2] <= 45.24108123779297) {
    if (x[1] <= 29.095568656921387) {
        if (x[0] <= 687.6794128417969) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 683.7836608886719) {
            votes[1] += 1;
        } else {

```

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        votes[0] += 1;
    }
}
} else {
    if (x[1] <= 18.206253051757812) {
        if (x[0] <= 684.3653259277344) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[1] <= 36.090810775756836) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}
// tree #5
if (x[2] <= 79.90730285644531) {
    if (x[1] <= 37.46395492553711) {
        if (x[0] <= 682.0504455566406) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 687.2783203125) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

    }
}
} else {
    if (x[0] <= 674.6184387207031) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
// tree #6
if (x[1] <= 18.463215827941895) {
    if (x[1] <= 18.42746353149414) {
        if (x[1] <= 18.05802059173584) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 19.531253814697266) {
        if (x[1] <= 19.27227020263672) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 38.38251876831055) {
            votes[0] += 1;
        } else {

```

```

        votes[1] += 1;
    }
}
}
// tree #7
if (x[1] <= 18.659820556640625) {
    if (x[2] <= 52.992692947387695) {
        if (x[2] <= 39.137678146362305) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[1] <= 18.11066436767578) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    if (x[2] <= 45.239837646484375) {
        if (x[1] <= 25.713486671447754) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[1] <= 20.5277042388916) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```

```

    }
}
}
// tree #8
if (x[0] <= 682.0504455566406) {
    if (x[1] <= 34.88931083679199) {
        votes[1] += 1;
    } else {
        if (x[2] <= 44.45821762084961) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}
// tree #9
if (x[2] <= 38.50187301635742) {
    if (x[1] <= 30.23251438140869) {
        if (x[0] <= 715.0144958496094) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 681.8073425292969) {
        if (x[1] <= 34.88931083679199) {

```

```

        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}
}
// tree #10
if (x[0] <= 683.3670654296875) {
    if (x[1] <= 34.89121437072754) {
        votes[1] += 1;
    } else {
        if (x[0] <= 462.7545928955078) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}
// tree #11
if (x[0] <= 682.0504455566406) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}
// tree #12
if (x[2] <= 79.99647903442383) {
    if (x[1] <= 18.23212718963623) {

```



```

    if (x[0] <= 653.3085632324219) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 18.42004680633545) {
        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
}
} else {
    if (x[2] <= 81.2431640625) {
        if (x[2] <= 81.07256317138672) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
}
}
// tree #13
if (x[2] <= 79.90730285644531) {
    if (x[0] <= 682.0504455566406) {
        if (x[0] <= 462.74522399902344) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```

```

    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 33.6096305847168) {
        if (x[1] <= 28.17322826385498) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[1] <= 36.94844436645508) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
// tree #14
if (x[2] <= 79.90730285644531) {
    if (x[0] <= 682.0504455566406) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 28.37818145751953) {
        if (x[0] <= 678.4080200195312) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

    }
} else {
    if (x[1] <= 29.748826026916504) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
}

```

// tree #15

```

if (x[1] <= 38.98320770263672) {
    if (x[1] <= 38.85782432556152) {
        if (x[0] <= 682.0504455566406) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[1] <= 38.929908752441406) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    }
} else {
    votes[1] += 1;
}
}

```

// tree #16

```

if (x[1] <= 21.970683097839355) {

```

```

if (x[2] <= 52.137779235839844) {
    if (x[0] <= 682.24755859375) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 60.09498977661133) {
        votes[0] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[0] <= 681.7316284179688) {
        if (x[2] <= 44.3887996673584) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
}
}

```

// tree #17

```

if (x[0] <= 682.0504455566406) {
    if (x[2] <= 44.373186111450195) {
        if (x[2] <= 44.348419189453125) {
            votes[1] += 1;
        } else {

```

```

        votes[0] += 1;
    }
} else {
    votes[1] += 1;
}
} else {
    votes[0] += 1;
}

```

// tree #18

```

if (x[0] <= 683.3670654296875) {
    if (x[1] <= 34.89121437072754) {
        votes[1] += 1;
    } else {
        if (x[1] <= 34.91578483581543) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}

```

// tree #19

```

if (x[1] <= 37.37936019897461) {
    if (x[0] <= 682.6343688964844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

} else {
    if (x[0] <= 684.3641357421875) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

// tree #20
if (x[2] <= 66.3783187866211) {
    if (x[2] <= 66.15283966064453) {
        if (x[1] <= 26.26651096343994) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 683.3496704101562) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
}

```

```

// tree #21
if (x[1] <= 37.40279769897461) {
    if (x[2] <= 81.12833404541016) {
        if (x[2] <= 38.52534103393555) {

```

```

        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
} else {
    if (x[0] <= 661.1711730957031) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[0] <= 687.2783203125) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
}

```

// tree #22

```

if (x[0] <= 682.3912658691406) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}
}

```

// tree #23

```

if (x[1] <= 29.071242332458496) {
    if (x[2] <= 45.85513114929199) {
        if (x[2] <= 44.784420013427734) {
            votes[1] += 1;
        }
    }
}
}

```

```

    } else {
        votes[1] += 1;
    }
} else {
    if (x[2] <= 46.45131874084473) {
        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
}
} else {
    if (x[1] <= 29.313894271850586) {
        if (x[1] <= 29.2493953704834) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[1] <= 29.637340545654297) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}

```

// tree #24

```

if (x[1] <= 19.158023834228516) {
    if (x[2] <= 55.6909122467041) {
        if (x[2] <= 48.64682388305664) {
            votes[0] += 1;
        }
    }
}

```



```

    } else {
        votes[1] += 1;
    }
} else {
    if (x[0] <= 691.6436462402344) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[2] <= 38.52828025817871) {
        if (x[0] <= 694.5677185058594) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[2] <= 40.70607948303223) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
}
}

```

// tree #25

```

if (x[1] <= 21.983495712280273) {
    if (x[2] <= 52.09625053405762) {
        if (x[0] <= 685.062255859375) {
            votes[1] += 1;
        }
    }
}

```

```

    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 684.1899719238281) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[0] <= 681.5479125976562) {
        if (x[1] <= 34.88931083679199) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
}
}

```

// tree #26

```

if (x[0] <= 682.3912658691406) {
    if (x[1] <= 34.88493728637695) {
        votes[1] += 1;
    } else {
        if (x[2] <= 44.53267860412598) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```

```

    }
}
} else {
    votes[0] += 1;
}

// tree #27
if (x[2] <= 45.24108123779297) {
    if (x[1] <= 18.058218955993652) {
        votes[0] += 1;
    } else {
        if (x[2] <= 43.97938537597656) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    if (x[0] <= 682.6343688964844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

// tree #28
if (x[1] <= 38.84697914123535) {
    if (x[1] <= 38.77797508239746) {
        if (x[0] <= 681.8073425292969) {
            votes[1] += 1;
        } else {

```

```

        votes[0] += 1;
    }
} else {
    votes[1] += 1;
}
} else {
    if (x[1] <= 38.98320770263672) {
        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
}

// tree #29
if (x[0] <= 681.8073425292969) {
    if (x[2] <= 44.373186111450195) {
        if (x[2] <= 44.348419189453125) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}

// tree #30
if (x[1] <= 19.642009735107422) {
    if (x[1] <= 19.550138473510742) {

```

```

    if (x[1] <= 18.00550937652588) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 19.576180458068848) {
        votes[0] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[2] <= 80.87892150878906) {
        if (x[1] <= 20.055273056030273) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[0] <= 660.0310974121094) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}

```

// tree #31

```

if (x[1] <= 38.74091148376465) {
    if (x[0] <= 682.0504455566406) {

```

```

    if (x[2] <= 44.373186111450195) {
        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}
} else {
    if (x[1] <= 38.98771667480469) {
        if (x[0] <= 676.7207336425781) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
}
}

```

// tree #32

```

if (x[2] <= 66.59783172607422) {
    if (x[1] <= 31.09712028503418) {
        if (x[2] <= 42.26988410949707) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 41.435922622680664) {
            votes[0] += 1;
        }
    }
}

```

```

    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[1] <= 37.40279769897461) {
        if (x[2] <= 77.53137588500977) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[2] <= 77.79978561401367) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}
}

```

// tree #33

```

if (x[0] <= 681.8073425292969) {
    if (x[2] <= 44.3887996673584) {
        if (x[2] <= 44.348419189453125) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
}
}

```

```
} else {  
    votes[0] += 1;  
}
```

```
// tree #34
```

```
if (x[0] <= 681.6485595703125) {  
    if (x[1] <= 34.89121437072754) {  
        votes[1] += 1;  
    } else {  
        if (x[0] <= 462.7545928955078) {  
            votes[1] += 1;  
        } else {  
            votes[1] += 1;  
        }  
    }  
} else {  
    votes[0] += 1;  
}
```

```
// tree #35
```

```
if (x[2] <= 79.90730285644531) {  
    if (x[1] <= 38.3791561126709) {  
        if (x[0] <= 682.6343688964844) {  
            votes[1] += 1;  
        } else {  
            votes[0] += 1;  
        }  
    } else {  
        if (x[2] <= 58.086130142211914) {  
            votes[0] += 1;  
        } else {
```



```

        votes[0] += 1;
    }
}
} else {
    if (x[0] <= 673.0796813964844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

// tree #36
if (x[0] <= 681.8073425292969) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

```

```

// tree #37
if (x[0] <= 682.0504455566406) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

```

```

// tree #38
if (x[1] <= 18.519329071044922) {
    if (x[0] <= 671.82373046875) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

    }
} else {
    if (x[0] <= 681.7316284179688) {
        if (x[0] <= 462.74522399902344) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
}

```

// tree #39

```

if (x[1] <= 37.17593002319336) {
    if (x[0] <= 681.8073425292969) {
        if (x[1] <= 34.89121437072754) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 38.094757080078125) {
        if (x[0] <= 673.6995544433594) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

    } else {
        if (x[2] <= 46.17604064941406) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```

```

// tree #40

```

```

if (x[1] <= 22.348251342773438) {
    if (x[0] <= 684.8724060058594) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 55.70467948913574) {
        if (x[0] <= 680.07958984375) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 682.5552368164062) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}

```

```

// tree #41
if (x[0] <= 682.3155517578125) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

// tree #42
if (x[2] <= 38.38251876831055) {
    if (x[2] <= 38.02525329589844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 49.377899169921875) {
        if (x[2] <= 48.9415397644043) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 51.32455635070801) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}
}

```

```

// tree #43
if (x[0] <= 682.3912658691406) {
    if (x[2] <= 44.373186111450195) {
        if (x[1] <= 34.69242858886719) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}

```

```

// tree #44
if (x[2] <= 62.13208770751953) {
    if (x[1] <= 29.75680923461914) {
        if (x[1] <= 28.22496509552002) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 38.467233657836914) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    }
} else {

```

```

if (x[2] <= 62.885345458984375) {
    if (x[2] <= 62.14506721496582) {
        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
} else {
    if (x[1] <= 38.095075607299805) {
        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
}
}

```

// tree #45

```

if (x[2] <= 71.19255447387695) {
    if (x[0] <= 681.6485595703125) {
        if (x[1] <= 34.88931083679199) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 681.7484741210938) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

    }
}

// tree #46
if (x[1] <= 38.54752159118652) {
    if (x[1] <= 38.409549713134766) {
        if (x[2] <= 80.8259048461914) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 649.65673828125) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
} else {
    if (x[1] <= 38.85782432556152) {
        if (x[2] <= 39.830522537231445) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[1] <= 38.98771667480469) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```

```

    }
}

// tree #47
if (x[1] <= 37.87756156921387) {
    if (x[1] <= 37.46395492553711) {
        if (x[1] <= 36.990861892700195) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 58.808115005493164) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    if (x[0] <= 687.3336791992188) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
}

```

```

// tree #48
if (x[0] <= 681.7316284179688) {
    if (x[0] <= 462.61668395996094) {
        if (x[0] <= 462.03025817871094) {
            votes[1] += 1;

```



```

    } else {
        votes[0] += 1;
    }
} else {
    votes[1] += 1;
}
} else {
    votes[0] += 1;
}

// tree #49
if (x[2] <= 77.00699615478516) {
    if (x[0] <= 682.0504455566406) {
        if (x[0] <= 462.61668395996094) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 79.1602897644043) {
        if (x[0] <= 678.93798828125) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 675.4946594238281) {
            votes[1] += 1;

```

```

    } else {
        votes[0] += 1;
    }
}

// tree #50
if (x[2] <= 81.09210205078125) {
    if (x[0] <= 682.6343688964844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 81.26008224487305) {
        if (x[0] <= 673.4417114257812) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
}

// tree #51
if (x[1] <= 38.86773681640625) {
    if (x[0] <= 682.0504455566406) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

    }
} else {
    if (x[2] <= 70.21031188964844) {
        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
}

// tree #52
if (x[1] <= 38.845359802246094) {
    if (x[1] <= 38.78886795043945) {
        if (x[2] <= 81.07256317138672) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
} else {
    if (x[2] <= 67.13547897338867) {
        if (x[1] <= 38.92056465148926) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[1] <= 38.96529579162598) {
            votes[1] += 1;
        } else {

```

```

        votes[0] += 1;
    }
}

// tree #53
if (x[0] <= 682.0504455566406) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

// tree #54
if (x[2] <= 81.0928840637207) {
    if (x[0] <= 682.79833984375) {
        if (x[0] <= 462.61668395996094) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 31.409253120422363) {
        votes[0] += 1;
    } else {
        if (x[1] <= 34.44680404663086) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```
    }  
  }  
}
```

```
// tree #55
```

```
if (x[2] <= 38.50187301635742) {  
  if (x[1] <= 28.005842208862305) {  
    if (x[0] <= 688.1924438476562) {  
      votes[1] += 1;  
    } else {  
      votes[0] += 1;  
    }  
  } else {  
    votes[0] += 1;  
  }  
} else {  
  if (x[0] <= 682.79833984375) {  
    if (x[1] <= 34.89121437072754) {  
      votes[1] += 1;  
    } else {  
      votes[1] += 1;  
    }  
  } else {  
    votes[0] += 1;  
  }  
}
```

```
// tree #56
```

```
if (x[1] <= 38.98005485534668) {  
  if (x[2] <= 81.07256317138672) {  
    if (x[1] <= 38.84697914123535) {
```

```

        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 661.1711730957031) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    votes[1] += 1;
}

```

// tree #57

```

if (x[0] <= 683.3222961425781) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

```

// tree #58

```

if (x[1] <= 20.44952964782715) {
    if (x[0] <= 684.4705200195312) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 681.8073425292969) {

```

```
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
```

```
// tree #59
```

```
if (x[0] <= 682.79833984375) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}
```

```
// tree #60
```

```
if (x[1] <= 18.38436508178711) {
    if (x[2] <= 52.271493911743164) {
        if (x[0] <= 671.8171081542969) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[1] <= 18.176514625549316) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    }
} else {
    if (x[1] <= 18.409366607666016) {
        votes[1] += 1;
    }
}
```

```

    } else {
        if (x[0] <= 682.0504455566406) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

// tree #61

```

if (x[1] <= 38.87688636779785) {
    if (x[0] <= 682.6343688964844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 647.9339294433594) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

// tree #62

```

if (x[0] <= 683.3670654296875) {
    if (x[2] <= 44.3887996673584) {
        if (x[0] <= 462.74522399902344) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```



```

    }
} else {
    votes[1] += 1;
}
} else {
    votes[0] += 1;
}

// tree #63
if (x[2] <= 80.27066040039062) {
    if (x[1] <= 37.44561767578125) {
        if (x[2] <= 75.02981948852539) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[1] <= 38.13470649719238) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    if (x[2] <= 80.47039413452148) {
        votes[0] += 1;
    } else {
        if (x[2] <= 80.78479766845703) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```
    }  
  }  
}
```

```
// tree #64
```

```
if (x[2] <= 66.38485717773438) {  
  if (x[1] <= 38.98005485534668) {  
    if (x[0] <= 682.0504455566406) {  
      votes[1] += 1;  
    } else {  
      votes[0] += 1;  
    }  
  } else {  
    votes[1] += 1;  
  }  
} else {  
  if (x[0] <= 683.3496704101562) {  
    votes[1] += 1;  
  } else {  
    votes[0] += 1;  
  }  
}
```

```
// tree #65
```

```
if (x[0] <= 682.0504455566406) {  
  if (x[2] <= 44.3887996673584) {  
    if (x[0] <= 462.61668395996094) {  
      votes[1] += 1;  
    } else {  
      votes[1] += 1;  
    }  
  }  
}
```

```

    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}

```

// tree #66

```

if (x[0] <= 682.0504455566406) {
    if (x[1] <= 34.89121437072754) {
        votes[1] += 1;
    } else {
        if (x[1] <= 34.92845344543457) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}

```

// tree #67

```

if (x[1] <= 21.983495712280273) {
    if (x[2] <= 45.39162063598633) {
        if (x[1] <= 20.699403762817383) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {

```

```

        if (x[0] <= 683.9784851074219) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
} else {
    if (x[1] <= 22.235862731933594) {
        if (x[1] <= 22.094776153564453) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[0] <= 681.8073425292969) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}

```

// tree #68

```

if (x[0] <= 681.8073425292969) {
    if (x[0] <= 462.74522399902344) {
        if (x[2] <= 44.39048957824707) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {

```

```

        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}

// tree #69
if (x[1] <= 18.463215827941895) {
    if (x[0] <= 671.8171081542969) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 80.87892150878906) {
        if (x[2] <= 80.49773406982422) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 81.26008224487305) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

// tree #70
if (x[1] <= 38.86773681640625) {

```

```

if (x[0] <= 682.0504455566406) {
    if (x[2] <= 44.373186111450195) {
        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}
} else {
    if (x[2] <= 51.726057052612305) {
        if (x[2] <= 47.15701103210449) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[0] <= 673.4459533691406) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}

```

// tree #71

```

if (x[0] <= 682.0504455566406) {
    if (x[0] <= 462.61668395996094) {
        if (x[0] <= 461.67144775390625) {
            votes[1] += 1;
        } else {

```

```

        votes[0] += 1;
    }
} else {
    votes[1] += 1;
}
} else {
    votes[0] += 1;
}

```

// tree #72

```

if (x[0] <= 681.8073425292969) {
    if (x[2] <= 44.3887996673584) {
        if (x[0] <= 462.61668395996094) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}

```

// tree #73

```

if (x[1] <= 18.05802059173584) {
    votes[0] += 1;
} else {
    if (x[1] <= 38.27148628234863) {
        if (x[2] <= 81.09210205078125) {
            votes[1] += 1;
        }
    }
}

```

```

    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 69.17631149291992) {
        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
}
}
}

```

// tree #74

```

if (x[0] <= 681.8073425292969) {
    if (x[1] <= 34.89121437072754) {
        votes[1] += 1;
    } else {
        if (x[1] <= 34.91188621520996) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}
}

```

// tree #75

```

if (x[0] <= 681.8073425292969) {
    if (x[1] <= 34.88493728637695) {
        votes[1] += 1;
    }
}
}

```



```

    } else {
        if (x[2] <= 44.45821762084961) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}

```

// tree #76

```

if (x[1] <= 18.055389404296875) {
    votes[0] += 1;
} else {
    if (x[0] <= 682.6343688964844) {
        if (x[2] <= 44.373186111450195) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
}

```

// tree #77

```

if (x[0] <= 682.3155517578125) {
    if (x[2] <= 44.3887996673584) {
        if (x[0] <= 462.61668395996094) {
            votes[1] += 1;
        }
    }
}

```

```

    } else {
        votes[1] += 1;
    }
} else {
    votes[1] += 1;
}
} else {
    votes[0] += 1;
}

// tree #78
if (x[0] <= 681.6485595703125) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

// tree #79
if (x[2] <= 71.19255447387695) {
    if (x[2] <= 54.69313430786133) {
        if (x[2] <= 54.56878662109375) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[0] <= 683.5653991699219) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

    }
} else {
    if (x[0] <= 682.51318359375) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

// tree #80
if (x[1] <= 19.77092456817627) {
    if (x[2] <= 52.19268798828125) {
        if (x[1] <= 18.041059494018555) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 78.31377792358398) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    }
} else {
    if (x[0] <= 681.8073425292969) {
        if (x[0] <= 462.61668395996094) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
}

```

```

    } else {
        votes[0] += 1;
    }
}

// tree #81
if (x[2] <= 38.50187301635742) {
    if (x[0] <= 694.5677185058594) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 38.54752159118652) {
        if (x[2] <= 77.00937271118164) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 693.0975036621094) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

// tree #82
if (x[2] <= 71.19255447387695) {
    if (x[1] <= 38.02694320678711) {

```

```

    if (x[0] <= 683.1239624023438) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 41.39290428161621) {
        votes[0] += 1;
    } else {
        votes[1] += 1;
    }
}
} else {
    if (x[0] <= 682.51318359375) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

// tree #83
if (x[0] <= 682.0504455566406) {
    if (x[2] <= 44.373186111450195) {
        if (x[2] <= 44.348419189453125) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
}

```

```

    } else {
        votes[0] += 1;
    }

// tree #84
if (x[0] <= 682.0504455566406) {
    votes[1] += 1;
} else {
    votes[0] += 1;
}

// tree #85
if (x[1] <= 18.463215827941895) {
    if (x[0] <= 670.6153869628906) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 39.42944145202637) {
        if (x[1] <= 37.61083793640137) {
            votes[0] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[0] <= 682.0504455566406) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}

```

```

    }
}

// tree #86
if (x[2] <= 81.09210205078125) {
    if (x[0] <= 682.0504455566406) {
        if (x[2] <= 44.373186111450195) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 532.0234832763672) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

```

```

// tree #87
if (x[1] <= 18.461002349853516) {
    if (x[0] <= 671.8171081542969) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 682.6343688964844) {

```

```

        if (x[1] <= 34.88931083679199) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
}

```

// tree #88

```

if (x[0] <= 682.5552368164062) {
    if (x[1] <= 34.89121437072754) {
        votes[1] += 1;
    } else {
        if (x[0] <= 462.7545928955078) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}

```

// tree #89

```

if (x[2] <= 38.42219543457031) {
    votes[0] += 1;
} else {
    if (x[1] <= 21.310211181640625) {
        if (x[1] <= 21.192763328552246) {

```



```

        votes[0] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 21.58940601348877) {
        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
}
}

```

// tree #90

```

if (x[0] <= 682.0504455566406) {
    if (x[0] <= 462.61668395996094) {
        if (x[0] <= 462.01026916503906) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}

```

// tree #91

```

if (x[1] <= 20.28431224822998) {
    if (x[2] <= 46.15955924987793) {

```

```

    if (x[1] <= 20.25676155090332) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 20.18445587158203) {
        votes[0] += 1;
    } else {
        votes[0] += 1;
    }
}
} else {
    if (x[0] <= 682.2729797363281) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}
}

// tree #92
if (x[1] <= 37.50950622558594) {
    if (x[1] <= 36.42031669616699) {
        if (x[1] <= 36.373802185058594) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        if (x[2] <= 51.07293701171875) {
            votes[0] += 1;
        }
    }
}

```

```

    } else {
        votes[1] += 1;
    }
}
} else {
    if (x[1] <= 37.614139556884766) {
        votes[0] += 1;
    } else {
        if (x[0] <= 684.3641357421875) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    }
}
}

```

```

// tree #93

```

```

if (x[0] <= 681.8073425292969) {
    if (x[2] <= 44.373186111450195) {
        if (x[2] <= 44.33450126647949) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}
}

```

```

// tree #94
if (x[2] <= 38.516231536865234) {
    if (x[0] <= 688.1924438476562) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 682.3912658691406) {
        if (x[1] <= 34.89121437072754) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        votes[0] += 1;
    }
}

```

```

// tree #95
if (x[1] <= 18.058218955993652) {
    if (x[0] <= 687.649658203125) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[2] <= 81.09210205078125) {
        if (x[1] <= 18.602431297302246) {
            votes[1] += 1;
        } else {

```

```

        votes[1] += 1;
    }
} else {
    votes[0] += 1;
}
}

```

// tree #96

```

if (x[0] <= 682.3155517578125) {
    if (x[1] <= 34.89121437072754) {
        votes[1] += 1;
    } else {
        if (x[1] <= 34.91578483581543) {
            votes[0] += 1;
        } else {
            votes[1] += 1;
        }
    }
} else {
    votes[0] += 1;
}

```

// tree #97

```

if (x[2] <= 38.432024002075195) {
    if (x[0] <= 694.5677185058594) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 19.77259635925293) {

```

```

    if (x[2] <= 42.431758880615234) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 20.05588150024414) {
        votes[1] += 1;
    } else {
        votes[1] += 1;
    }
}
}

```

// tree #98

```

if (x[1] <= 19.79801082611084) {
    if (x[0] <= 687.4760437011719) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
} else {
    if (x[1] <= 19.9541015625) {
        if (x[1] <= 19.84791660308838) {
            votes[1] += 1;
        } else {
            votes[1] += 1;
        }
    } else {
        if (x[2] <= 38.52534103393555) {
            votes[0] += 1;
        }
    }
}

```

```

    } else {
        votes[1] += 1;
    }
}
}

// tree #99
if (x[2] <= 38.5191707611084) {
    if (x[1] <= 29.60585308074951) {
        if (x[0] <= 688.1924438476562) {
            votes[1] += 1;
        } else {
            votes[0] += 1;
        }
    } else {
        votes[0] += 1;
    }
} else {
    if (x[0] <= 682.6343688964844) {
        votes[1] += 1;
    } else {
        votes[0] += 1;
    }
}

// tree #100
if (x[0] <= 682.0504455566406) {
    if (x[0] <= 462.74522399902344) {
        if (x[2] <= 44.41454315185547) {
            votes[1] += 1;
        } else {

```

```

        votes[1] += 1;
    }
} else {
    votes[1] += 1;
}
} else {
    votes[0] += 1;
}

// return argmax of votes
uint8_t classIdx = 0;
float maxVotes = votes[0];

for (uint8_t i = 1; i < 2; i++) {
    if (votes[i] > maxVotes) {
        classIdx = i;
        maxVotes = votes[i];
    }
}
return classIdx;
}

protected:
};
}

```